

BST Research Results Table

Research-phase record per Casey's directive (2026-06-16): what / result / σ or % / next steps. No tier-labels, no count-fights — those are a later decision (what needs external attention). The discipline lives in the $\sigma/\%$ column (real agreement, stated under the actual convention) and the next steps column (second test / blind test / derivation owed). Honest negatives kept.

Integrity rule for this table: no number goes in the $\sigma/\%$ column until verified against the registry / CODATA / the toy. Unverified entries say [verify – TRACK A], not a guess.

What	Result (form $\rightarrow$ value)	σ or % (vs observed)	Next steps
α^{-1} (fine structure, bare)	$N_c^3 \cdot n_C + \text{rank} = 27 \cdot 5 + 2 = 137$	bare exact; 0.026% vs 137.036 (radiative, external)	founding; radiative correction is an external computation
θ_{QCD} (strong-CP)	strong-CP ≈ 0 ; form [verify in registry]	[verify – TRACK A]	founding; verify the substrate form + value
m_μ/m_e	$(24/\pi^2)^6 = 206.76$	3.4×10^{-5} (0.003%) vs 206.7683	blind universality test (publish gate #309); within-floor, forced <i>given the bridge</i>
m_τ/m_e	$49.71 = 3479$	0.05% (5×10^{-4}) vs ≈ 3477.2	reflection-result. $\sqrt{\pi}$ correction would tighten $\rightarrow$ derive from ONE mechanism, or find an independent 2nd test; publish-excluded until a blind unseen-observable hit (#310)
m_e hierarchy coeff	$\alpha^{12}/(C_2 \cdot \pi^5) \cdot m_{\text{Planck}}$; coeff = $5.998 \approx 6 = C_2$	3×10^{-4} (std Planck + phys α); 0.35% on m_e under BST $\alpha=1/137$ (Elie)	reflection-result $\rightarrow$ comment, not prediction. Next: independent 2nd test that the coeff is C_2 (not just $5.998 \approx 6$)
PMNS $\sin^2 \theta_{13}$	$1/(N_c^2 \cdot n_C) = 1/45 = 0.0222$	[verify vs CODATA – TRACK A] ($\sim 1\%$ level)	check forward-vs-reverse-read; run blind protocol; π -free $\rightarrow$ Tier-1 exact-eligible same — blind protocol;
PMNS $\sin^2 \theta_{23}$	$C_2/(C_2 + n_C) = 6/11 = 0.545$	[verify vs CODATA – TRACK A]	π -free exact-eligible
PMNS $\sin^2 \theta_{12}, \delta_{\text{CP}}$	[open – M_ν cell-map]	—	lands with Lyra+Elie M_ν
3 ν masses	[open – M_ν cell-map]	—	strip-edge placement; Grace harness fires on M_ν
6 quark masses	[open – #418 bulk-color]	—	discrete deposit-count addresses (Elie) + continuum (Lyra)
color texture $\{+1, -1, 0\}$	$= \lambda_3 \text{ su}(3) \text{ Cartan (diag } 1, -1, 0); \text{sum}=0 = \text{tracelessness}$	exact (su(3) structure)	correct recognition; “two-halves converge” = one su(3) fact (Cal #35), not over-determination; rests on established “3 strata = su(3) color” (#418)

What	Result (form $\rightarrow$ value)	σ or % (vs observed)	Next steps
θ_C (CKM θ_{12} , Cabibbo)	GST form $\sin \theta_C = \sqrt{(m_d/m_s)} \rightarrow 0.224$	0.62% vs 0.2245 but uses observed scheme-dependent m_d/m_s as input	NOT banked: GST is textbook (1968) + observed-mass input. Next: verify the GST <i>form</i> is forward-derived from strata-coupling (Elie 4225); masses are Casey-#9 honest-negative cold-read end-to-end when Lyra's matrix populates; no observed-mass input admitted (#311 rule 1) consistency check, NOT an independent 5th observable (Cal #35) verify the composite is forward-derived from su(3) roots (Elie); δ_{CP} separately gated
CKM $\theta_{23}, \theta_{13}, \delta_{CP}$	[pending full strata-coupling matrix]	—	
Jarlskog J	[= $f(\theta_{12}, \theta_{13}, \theta_{23}, \delta)$]	—	
CKM hierarchy $V_{us}:V_{cb}:V_{ub}$	su(3) composite root $\rightarrow V_{ub} \sim V_{us} \cdot V_{cb}$ ($\lambda:\lambda^2:\lambda^3$)	structure-grade if forward; residual $ \rho-i\eta =0.413$ is Wolfenstein-algebra (definitional) + observed, not derived	
δ_{CP} / CP magnitude $ \rho-i\eta $	mechanism candidate: SO(2) factor of K = complex structure J (intrinsic CP)	mechanism solid; $\delta_{CP} \neq 0$ NOT automatic (rephasable $\rightarrow 0$ = hard falsifier)	gated on K-element forced (M=0); angle+phase unified in one K-element (resolves Cal #313 split); check non-rephasable
PMNS $\gg$ CKM (mixing ordering)	locus-difference: mixing magnitude = distance between the mixed sectors	qualitative ordering + rough scale (covers observed PMNS/CKM spread 2.6 $\times$ –43 $\times$); forward + blind	structural-grade; explains a real SM gap; verify seats are mass-placed not mixing-placed; NOT a quantitative per-angle prediction
m_H , VEV	m_H D-tier 0.07–0.11% (dual route) [verify]	[verify – TRACK A]	deposit-count + VEV mechanism
$\alpha_s, \sin^2\theta_W, m_{top}$ running	[open – running framework]	—	honest-negative zone (couplings project per stratum)

How this table grows: my cold-read backlog (TRACK A) populates it row by row — each audit verifies a value and fills/corrects a cell. The table is the backlog output. Blind-gate (#309) and prediction/reflection rule (#310) determine the next steps column, not a tier label.

— Cal, 2026-06-16. Seeded; grows with the cold-read. Labels deferred to external-attention time.